

FINGERPRINT BASED ELECTRONIC VOTING MACHINE

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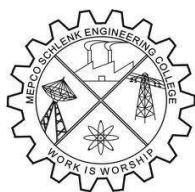
ELECTRONICS AND COMMUNICATION ENGINEERING

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in

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**DEPARTMENT OF ELECTRONICS
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MEPCO SCHLENK ENGINEERING COLLEGE, SIVAKASI
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CERTIFICATE

This is to certify that it is the Bonafide work done by Mr. AJAY P (9517202403007), Mr. VIJAY SRIDHAR P (9517202403181) for the 23EC353 - Mini Project - II work entitled Fingerprint Based Electronic Voting Machine at Department of Electronics and Communication Engineering, Mepco Schlenk Engineering College, Sivakasi during the year 2025-2026.

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Internal Examiner - I

Internal Examiner - II

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ABSTRACT

The project titled “Fingerprint Based Electronic Voting Machine” presents an advanced, secure, and user-centric prototype designed to address the critical challenges of voter identity theft and fraudulent voting. This system leverages biometric technology to ensure a "one person, one vote" policy by using a fingerprint sensor as the primary authentication interface.

In this design, the system utilizes a microcontroller— an Arduino interfaced with an optical fingerprint module to enroll and verify voters. When a user places their finger on the scanner, the system compares the live print against a pre-registered database. Once a vote is recorded, the system automatically locks the identity to prevent multiple entries by the same individual, storing the final tallies in the microcontroller’s non-volatile memory (EEPROM) to ensure data retention even during power failures.

By integrating biometric security, this project significantly enhances the integrity of the electoral process compared to traditional manual or basic electronic systems. Ultimately, the Fingerprint Based Electronic Voting Machine serves as a sophisticated model for modernizing democratic exercises, demonstrating the practical application of biometric sensors and embedded systems in creating a transparent, tamper-proof voting environment.

INTRODUCTION TO ELECTRONIC VOTING MACHINE-EVM's

Electronic Voting Machines (EVMs) represent a significant technological advancement in the process of conducting elections. Designed to replace traditional paper-based voting systems, aim to EVMs enhance the efficiency, accuracy, and reliability of the voting and vote-counting process. At their core, EVMs are electronic devices that allow voters to cast their ballots through a simple interface, typically involving a control unit operated by election officials and a balloting unit used by voters. This digital process not only streamlines the overall conduct of elections but also minimizes many common issues associated with manual voting methods.

Historically, paper ballots were prone to errors, manipulation, invalid votes, and lengthy counting procedures. These challenges often resulted in delays, disputes, and high administrative costs. Electronic Voting Machines were developed to overcome such limitations by ensuring greater security, reducing human intervention, and promoting transparency. Through features such as one vote per activation, secure internal memory, and tamper-resistant hardware, EVMs have become a trusted tool in many electoral systems around the world. They have been especially valuable in regions with large populations, where rapid counting and error reduction are essential.

The Arduino-based design makes the system cost-effective, simple, and easy to implement for small-scale applications. This approach ensures higher accuracy, prevents unauthorized access, and improves transparency in the voting process.

DIFFERENT TYPES OF EVM'S

Electronic Voting Machines have evolved over time, and several types are used worldwide depending on technological needs, security requirements, and the scale of elections. One common type is the Direct Recording Electronic (DRE) EVM, where voters cast their ballots using a touchscreen or push-button interface, and votes are recorded directly into electronic memory. Some DRE systems include a Voter Verifiable Paper Audit Trail (VVPAT), which provides a printed record for verification and auditing.

Another widely used type is the **Ballot Unit and Control Unit-based EVM**, commonly deployed in countries like India. This system uses a simple push-button interface on the Ballot Unit connected to a Control Unit that securely stores vote data.

More advanced systems include **optical scan voting machines**, which allow voters to mark paper ballots that are then scanned electronically for counting. There are also **internet-based or remote voting systems**, still emerging, which enable voting over secure digital platforms for specific cases such as overseas or disabled voters.

ADVANTAGES

- Since fingerprints are unique to every individual, it prevents impersonation and duplicate voting.
- Biometric authentication ensures that only registered voters can cast their vote.
- Reduces errors compared to manual ID verification methods like voter ID cards.

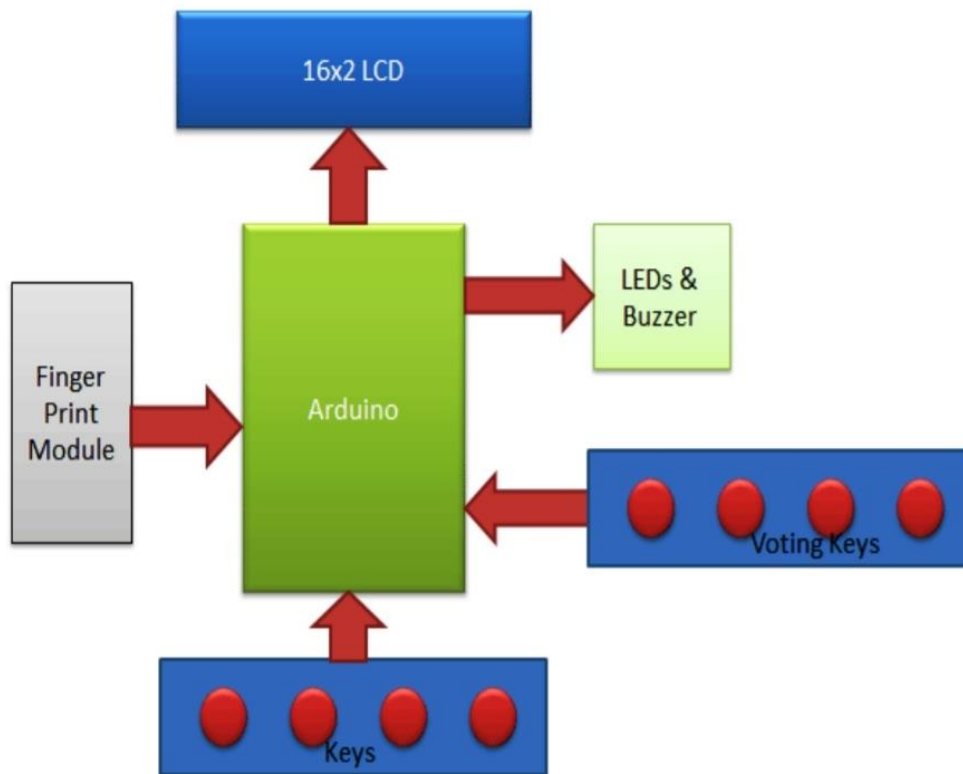
DISADVANTAGES

- Biometric sensors, storage systems, and software make the setup expensive.
- People with worn, damaged, or unclear fingerprints (elderly, laborers) may face authentication failure.
- Requires continuous electricity; power failure can halt voting.

APPLICATIONS

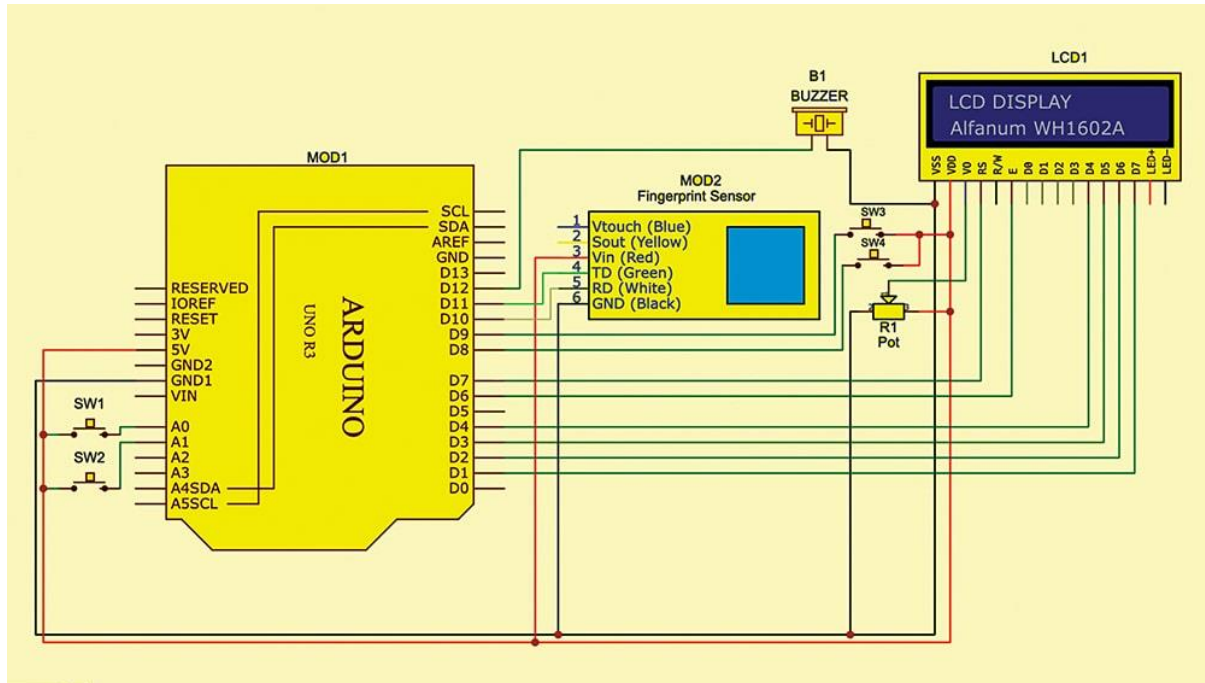
- National and Regional Elections: Widely used in general elections to record votes electronically, ensuring faster and more accurate counting of votes.
- Polling for Referendums: Used to conduct referendums, enabling quick vote tallying and transparent results.
- By-elections: Deployed to fill vacant legislative or parliamentary seats between general elections, improving efficiency and maintaining electoral integrity.
- Corporate or Organizational Voting: Employed in private sector elections, such as for board appointments or shareholder voting, ensuring security and confidentiality.
- E-Voting for Overseas Citizens: Facilitates voting for citizens living abroad, using secure remote voting methods to participate in national election.

BLOCK SCHEMATIC



This block diagram illustrates a fingerprint-based electronic voting system. The fingerprint module acts as the input device to authenticate voters by verifying their identity before allowing access to the voting process. Once verified, the user can cast their vote using the provided voting keys, while additional control keys are used for system operations such as reset or mode selection. The Arduino processes all inputs and controls the outputs, displaying relevant information like instructions and voting status on the 16×2 LCD. LEDs and a buzzer provide visual and audio feedback to indicate successful authentication, vote casting, or errors and authorized users to participate.

IMPLEMENTED CIRCUIT

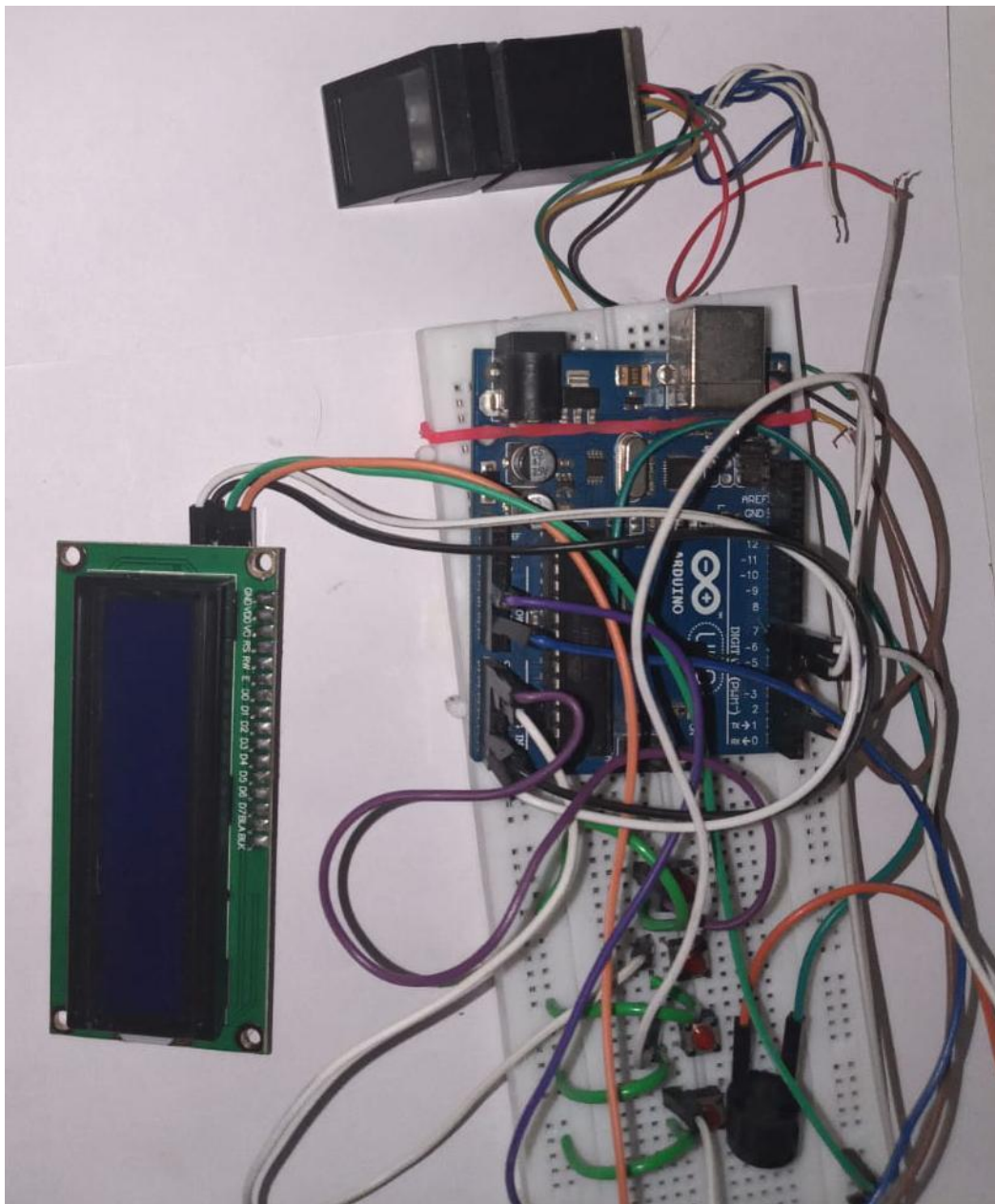


This circuit represents a fingerprint-based security system using an Arduino UNO, where the Arduino acts as the central controller connecting all components. A fingerprint sensor is interfaced with the Arduino to capture and verify the user's fingerprint data through serial communication. A 16×2 LCD display is used to show system messages such as prompts, authentication status, and instructions, while a potentiometer adjusts the LCD contrast for clear visibility. Push buttons are included for functions like enrolling and managing fingerprint data. Additionally, a buzzer provides audible feedback to indicate whether access is granted or denied. When a finger is placed on the sensor, the Arduino processes the input and compares it with stored templates; based on the result, it displays the appropriate message on the LCD and activates the buzzer.

CIRCUIT CONNECTIONS

1.BREABBOARD IMPLEMENTATION

The entire circuit is built on interconnected breadboards, using multiple IC CD4026 counters, seven-segment displays, and push-button inputs. The layout demonstrates a fully functional voting system prototype assembled purely through breadboard wiring and discrete components.



CURRENT EVM USED IN INDIA



The Electronic Voting Machine (EVM) used in India is a reliable and tamper-proof device designed by the Election Commission of India to ensure secure and transparent voting. It consists of two main units: the Ballot Unit and the Control Unit, connected through a cable. The Ballot Unit contains candidate names and symbols, allowing voters to cast their vote by pressing a single button. The Control Unit is operated only by authorized polling officers and records each vote securely. Indian EVMs are stand-alone, battery-operated devices with no internet or wireless connectivity, eliminating hacking risks. They use advanced microcontrollers and encrypted data storage for high security. Each vote is stored electronically and can be verified using VVPAT, which provides a printed slip visible to the voter for a few seconds. These machines support multi-candidate elections and ensure quick, error-free counting. The EVM system has improved efficiency, reduced invalid votes, and strengthened the integrity of the electoral process in India.

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CONCLUSION

In conclusion, Electronic Voting Machines (EVMs) represent a transformative step forward in the modernization of electoral processes. They offer significant improvements over traditional paper-based voting systems by ensuring faster, more accurate vote recording, reducing human error, and minimizing the risk of fraudulent activities. EVMs have not only enhanced the efficiency and security of elections but have also made the electoral process more transparent and accessible to voters

APPENDIX

ARDUINO PROGRAM

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Adafruit_Fingerprint.h>
#include <SoftwareSerial.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Fingerprint
SoftwareSerial mySerial(2, 3);
Adafruit_Fingerprint finger =
Adafruit_Fingerprint(&mySerial);

// Pins
#define SW1 A0 // Show Results
#define SW2 A1 // Reset Votes
#define SW3 6  // Vote A
#define SW4 7  // Vote B
#define BUZZER 8

// Votes
int votesA = 0;
int votesB = 0;

#define MAX_USERS 50
bool voted[MAX_USERS];
```

```
void setup() {  
  lcd.init();  
  lcd.backlight();  
  
  pinMode(SW1, INPUT_PULLUP);  
  pinMode(SW2, INPUT_PULLUP);  
  pinMode(SW3, INPUT_PULLUP);  
  pinMode(SW4, INPUT_PULLUP);  
  pinMode(BUZZER, OUTPUT);  
  
  Serial.begin(9600);  
  mySerial.begin(57600);  
  
  lcd.print("Voting System");  
  delay(2000);  
  
  if (!finger.verifyPassword()) {  
    lcd.clear();  
    lcd.print("Sensor Error!");  
    while (1);  
  }  
  
  lcd.clear();  
  lcd.print("Place Finger");  
}  
  
void loop() {
```

```
// Show Results
if (digitalRead(SW1) == LOW) {
    showResults();
}

// Reset Votes
if (digitalRead(SW2) == LOW) {
    resetVotes();
}

int id = getFingerprintID();

if (id != -1 && id < MAX_USERS) {

    lcd.clear();
    lcd.print("ID: ");
    lcd.print(id);
    delay(1000);

    if (voted[id]) {
        lcd.clear();
        lcd.print("Already Voted!");
        beep(2);
        delay(2000);
    } else {
        lcd.clear();
        lcd.print("Vote Now");
    }
}
```

```
int choice = waitForVote();

if (choice == 1) {
    votesA++;
    lcd.clear();
    lcd.print("Voted A");
}
else if (choice == 2) {
    votesB++;
    lcd.clear();
    lcd.print("Voted B");
}

voted[id] = true;
beep(1);
delay(2000);
}

lcd.clear();
lcd.print("Place Finger");
}

// Fingerprint
int getFingerprintID() {
    if (finger.getImage() != FINGERPRINT_OK) return -
1;
    if (finger.image2Tz() != FINGERPRINT_OK) return -
```

```
1;
  if (finger.fingerSearch() != FINGERPRINT_OK)
return -1;
```

```
    return finger.fingerID;
}
```

```
// Vote buttons
```

```
int waitForVote() {
  while (1) {
    if (digitalRead(SW3) == LOW) {
      delay(300);
      return 1;
    }
    if (digitalRead(SW4) == LOW) {
      delay(300);
      return 2;
    }
  }
}
```

```
// Show Results
```

```
void showResults() {
  lcd.clear();
  lcd.print("A:");
  lcd.print(votesA);
  lcd.setCursor(0,1);
  lcd.print("B:");
```

```
lcd.print(votesB);

beep(1);
delay(3000);

lcd.clear();
lcd.print("Place Finger");
}

// Reset Votes
void resetVotes() {
  votesA = 0;
  votesB = 0;

  for (int i = 0; i < MAX_USERS; i++) {
    voted[i] = false;
  }

  lcd.clear();
  lcd.print("Votes Reset");
  beep(2);
  delay(2000);

  lcd.clear();
  lcd.print("Place Finger");
}

// Buzzer
```

```
void beep(int times) {  
  for (int i = 0; i < times; i++) {  
    digitalWrite(BUZZER, HIGH);  
    delay(200);  
    digitalWrite(BUZZER, LOW);  
    delay(200);  
  }  
}
```